

Tecnológico de Monterrey

Materia:
Integración de seguridad informática en redes y sistemas de software (Gpo 601)

Actividad evaluativa:
4.4.9 Lab - Troubleshoot Inter-VLAN Routing

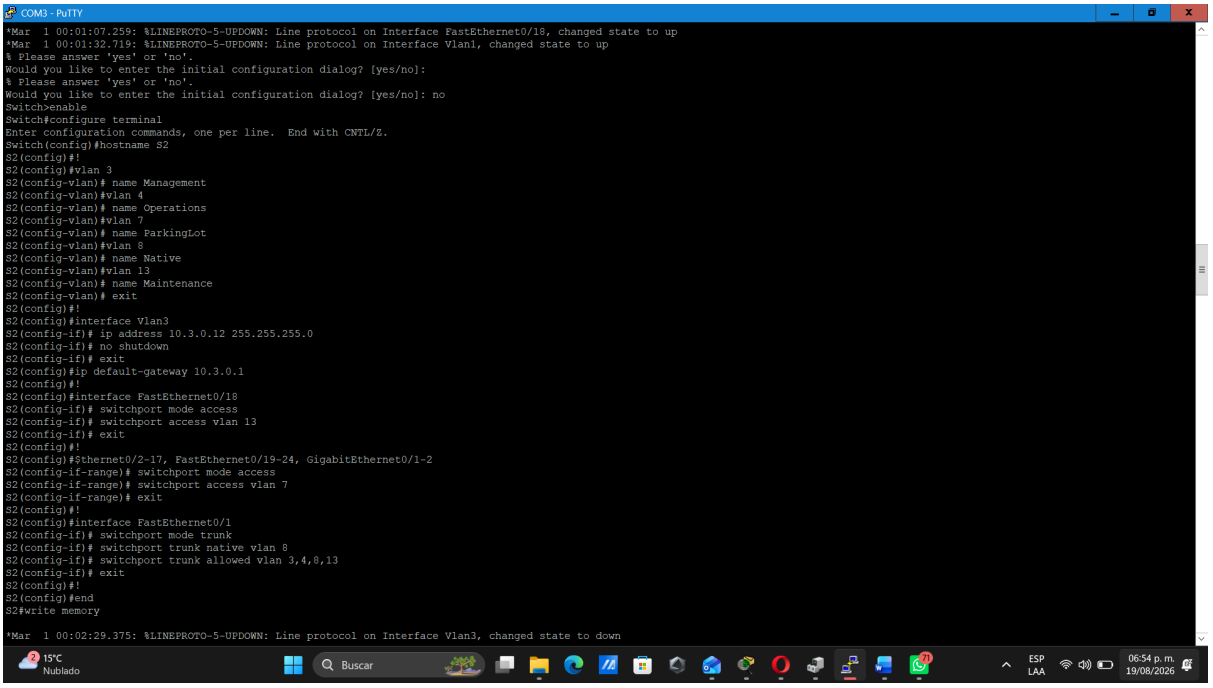
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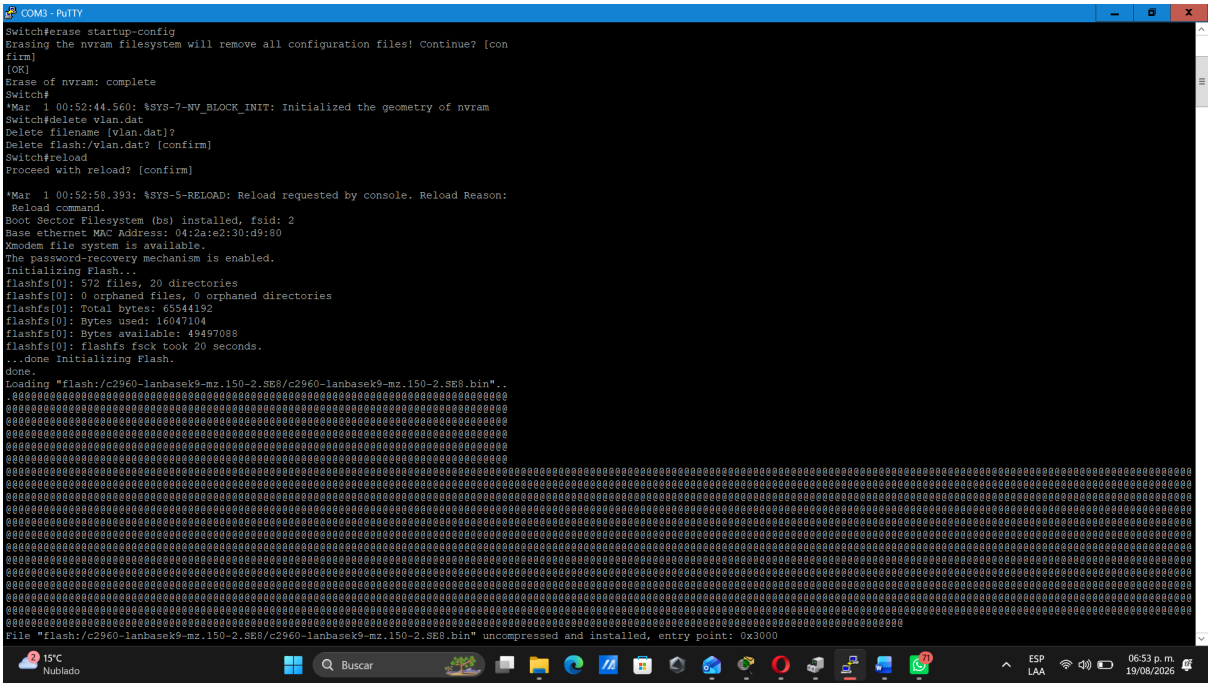
Fecha:
19 de agosto de 2026

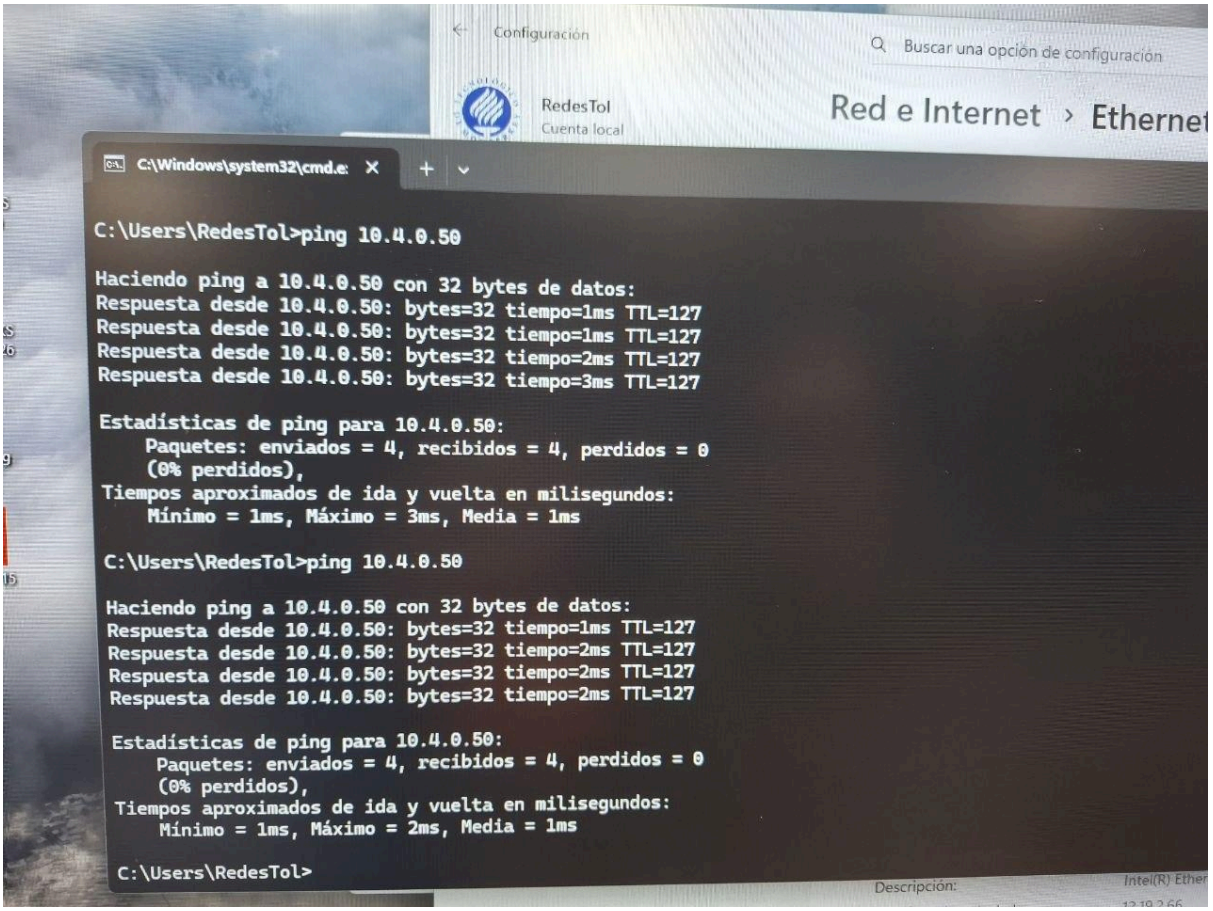
From	To	Ping Results
R1	S1 VLAN 3 (10.3.0.11)	
	S2 VLAN 3 (10.3.0.12)	
	PC-A (10.4.0.50)	
	PC-B (10.13.0.50)	
S1	S2 VLAN 3 (10.3.0.12)	
	PC-A (10.4.0.50)	
	PC-B (10.13.0.50)	
S2	PC-A (10.4.0.50)	
	PC-B (10.13.0.50)	

Configuración Switch 2



Reset del switch 2


COM3 - PuTTY
Switch#erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [con
firm]
[OK]
Erase of nvram: complete
Switch#
*Mar 1 00:52:44.560: %SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
Switch#delete vlan.dat
Delete filename [vlan.dat]?
Delete flash:/vlan.dat? [confirm]
Switch#reload
Proceed with reload? [confirm]
*Mar 1 00:52:58.393: %SYS-5-RELOAD: Reload requested by console. Reload Reason:
 Reload command.
Boot Sector Filesystem (bs) installed, fsid: 2
Base ethernet MAC Address: 04:2a:e2:30:d9:80
Xmodem file system is available.
The password-recovery mechanism is enabled.
Initializing Flash...
flashfs[0]: 572 files, 20 directories
flashfs[0]: 0 orphaned files, 0 orphaned directories
flashfs[0]: Total bytes: 65544192
flashfs[0]: Bytes used: 16047104
flashfs[0]: Bytes available: 49497088
flashfs[0]: flashfs fsck took 20 seconds.
...done Initializing Flash.
done.
Loading "flash:/c2960-lanbasek9-mz.150-2.SE8/c2960-lanbasek9-mz.150-2.SE8.bin"..
File "flash:/c2960-lanbasek9-mz.150-2.SE8/c2960-lanbasek9-mz.150-2.SE8.bin" uncompressed and installed, entry point: 0x3000
15°C
Nublado
Buscar
ESP
LAA
06:53 p. m.
19/08/2026


Configuración
Buscar una opción de configuración
RedesTol
Cuenta local
Red e Internet › Ethernet
C:\Windows\system32\cmd.e
C:\Users\RedesTol>ping 10.4.0.50
Haciendo ping a 10.4.0.50 con 32 bytes de datos:
Respuesta desde 10.4.0.50: bytes=32 tiempo=1ms TTL=127
Respuesta desde 10.4.0.50: bytes=32 tiempo=1ms TTL=127
Respuesta desde 10.4.0.50: bytes=32 tiempo=2ms TTL=127
Respuesta desde 10.4.0.50: bytes=32 tiempo=3ms TTL=127
Estadísticas de ping para 10.4.0.50:
Paquetes: enviados = 4, recibidos = 4, perdidos = 0
(0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
Mínimo = 1ms, Máximo = 3ms, Media = 1ms
C:\Users\RedesTol>ping 10.4.0.50
Haciendo ping a 10.4.0.50 con 32 bytes de datos:
Respuesta desde 10.4.0.50: bytes=32 tiempo=1ms TTL=127
Respuesta desde 10.4.0.50: bytes=32 tiempo=2ms TTL=127
Respuesta desde 10.4.0.50: bytes=32 tiempo=2ms TTL=127
Respuesta desde 10.4.0.50: bytes=32 tiempo=2ms TTL=127
Estadísticas de ping para 10.4.0.50:
Paquetes: enviados = 4, recibidos = 4, perdidos = 0
(0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
Mínimo = 1ms, Máximo = 2ms, Media = 1ms
C:\Users\RedesTol>
Descripción:

Ping de PC-A

```
Ping statistics for 10.13.0.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Users\salin>ping 10.13.0.50

Pinging 10.13.0.50 with 32 bytes of data:
Reply from 10.13.0.50: bytes=32 time=2ms TTL=127
Reply from 10.13.0.50: bytes=32 time=2ms TTL=127
Reply from 10.13.0.50: bytes=32 time=3ms TTL=127
Reply from 10.13.0.50: bytes=32 time=2ms TTL=127

Ping statistics for 10.13.0.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

```
C:\Users\salin>ping 10.3.0.12

Pinging 10.3.0.12 with 32 bytes of data:
Reply from 10.3.0.12: bytes=32 time=2ms TTL=254
Reply from 10.3.0.12: bytes=32 time=6ms TTL=254
Reply from 10.3.0.12: bytes=32 time=2ms TTL=254
Reply from 10.3.0.12: bytes=32 time=2ms TTL=254

Ping statistics for 10.3.0.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 6ms, Average = 3ms

C:\Users\salin>ping 10.3.0.11

Pinging 10.3.0.11 with 32 bytes of data:
Reply from 10.3.0.11: bytes=32 time=3ms TTL=253
Reply from 10.3.0.11: bytes=32 time=3ms TTL=253
Reply from 10.3.0.11: bytes=32 time=3ms TTL=253
Reply from 10.3.0.11: bytes=32 time=3ms TTL=253

Ping statistics for 10.3.0.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 3ms, Average = 3ms
```

Configuration Switch 1

```
*Aug 19 23:12:29.100: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet2/0/1 (8), with S2 FastEthernet0/1 (1000).
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#
S1(config)#interface GigabitEthernet2/0/5
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 8
S1(config-if)# switchport trunk allowed vlan 3,4,8,13
S1(config-if)# no shutdown
S1(config-if)#exit
S1(config)#
S1(config)#end
S1#copy running-config startup-config
*Aug 19 23:12:35.215: %SYS-5-CONFIG_I: Configured from console by console
Destination filename [startup-config]?
Building configuration...
[OK]
S1#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Gi2/0/1     on               802.1q         trunking      8

Port        Vlans allowed on trunk
Gi2/0/1     3-4,8,13

Port        Vlans allowed and active in management domain
Gi2/0/1     3-4,8,13

Port        Vlans in spanning tree forwarding state and not pruned
Gi2/0/1     3-4,13
S1#
S1#
*Aug 19 23:13:29.105: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet2/0/1 (8), with S2 FastEthernet0/1 (1000).
S1#show interfaces GigabitEthernet2/0/5 status

Port      Name               Status       Vlan       Duplex  Speed Type
Gi2/0/5                      notconnect   1            auto   auto 10/100/1000BaseTX
```